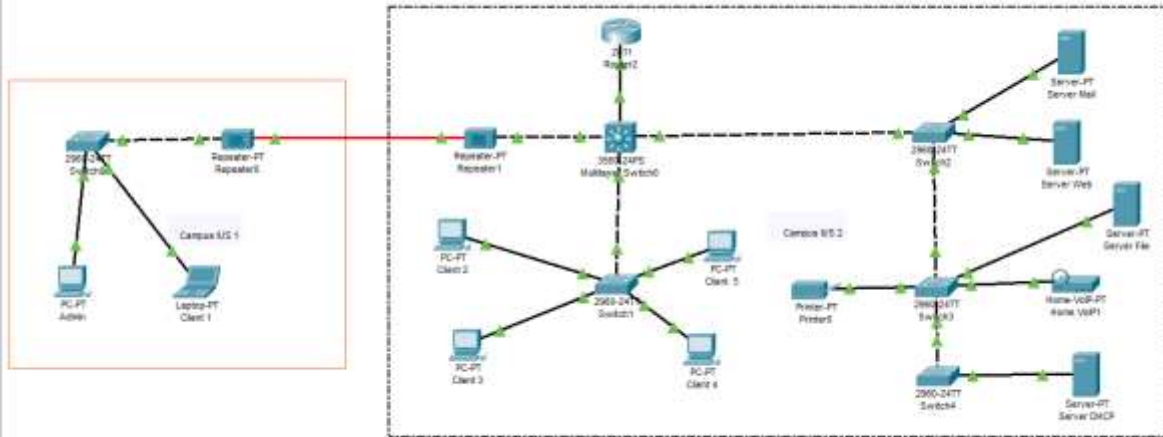
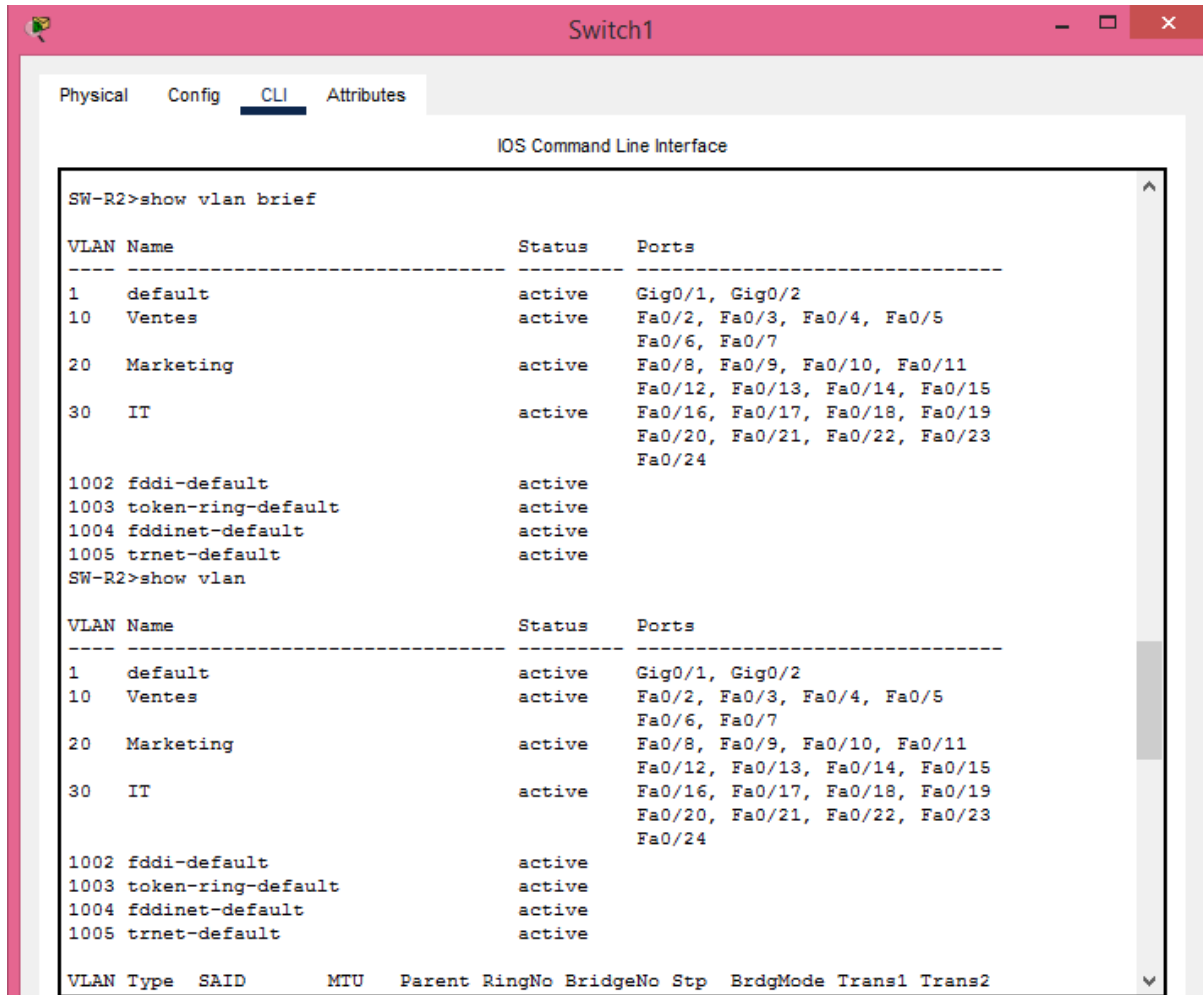


The diagram illustrates a multi-site network topology. A central 'Campus B/S 1' (dashed box) is connected to 'Campus B/S 2' (dashed box). Campus B/S 1 includes a 2960-24TT Switch, a 2960-24TT Multiswitch, and a 2960-24TT Switch. Campus B/S 2 includes a 2960-24TT Switch, a 2960-24TT Switch, and a 2960-24TT Switch. Various servers and clients are connected to these switches.



Vérifications importantes

- Vérification des vlans sur :



The screenshot shows a network switch configuration window titled "Switch1". The "CLI" tab is selected, displaying the "IOS Command Line Interface". The command "SW-R2>show vlan brief" has been executed, resulting in two identical tables showing the configuration of VLANs 1, 10, 20, and 30, along with their associated ports and status.

VLAN	Name	Status	Ports
1	default	active	Gig0/1, Gig0/2
10	Ventes	active	Fa0/2, Fa0/3, Fa0/4, Fa0/5 Fa0/6, Fa0/7
20	Marketing	active	Fa0/8, Fa0/9, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15
30	IT	active	Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24

VLAN	Name	Status	Ports
1	default	active	Gig0/1, Gig0/2
10	Ventes	active	Fa0/2, Fa0/3, Fa0/4, Fa0/5 Fa0/6, Fa0/7
20	Marketing	active	Fa0/8, Fa0/9, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15
30	IT	active	Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24

Below the tables, the following VLANs are listed as active:

- 1002 fddi-default
- 1003 token-ring-default
- 1004 fddinet-default
- 1005 trnet-default

At the bottom of the window, a header for a detailed VLAN table is visible:

VLAN	Type	SAID	MTU	Parent	RingNo	BridgeNo	Stp	BrdgMode	Trans1	Trans2
------	------	------	-----	--------	--------	----------	-----	----------	--------	--------

Switch2

PhysicalConfigCLIAttributes

IOS Command Line Interface

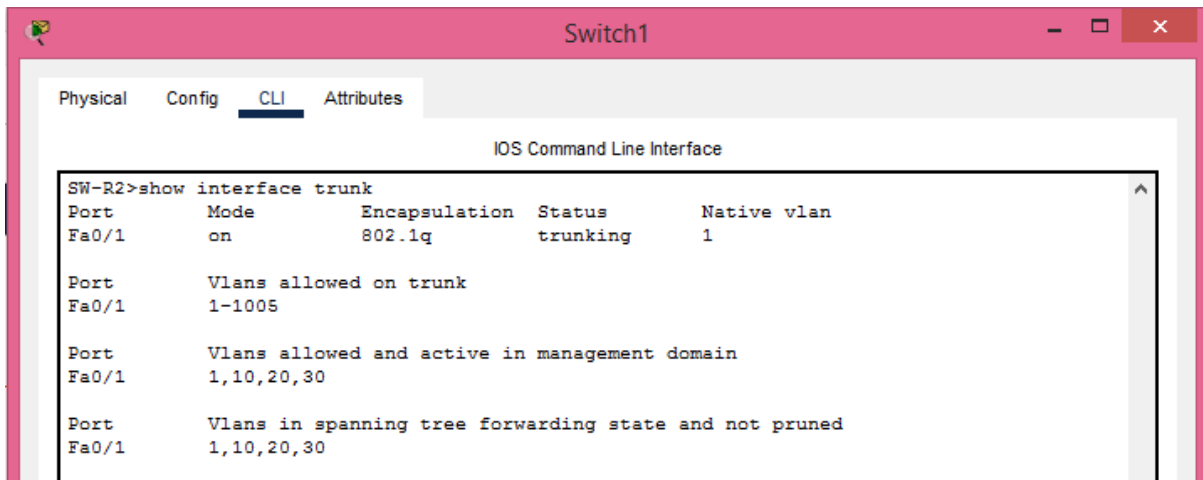
VLAN	Name	Status	Ports
1	default	active	Gig0/1, Gig0/2
10	Ventes	active	Fa0/2, Fa0/3, Fa0/4, Fa0/5 Fa0/6, Fa0/7
20	Marketing	active	Fa0/8, Fa0/9, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15
30	IT	active	Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

SW-R2>show vlan

VLAN	Name	Status	Ports
1	default	active	Gig0/1, Gig0/2
10	Ventes	active	Fa0/2, Fa0/3, Fa0/4, Fa0/5 Fa0/6, Fa0/7
20	Marketing	active	Fa0/8, Fa0/9, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15
30	IT	active	Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

VLAN	Type	SAID	MTU	Parent	RingNo	BridgeNo	Stp	BrdgMode	Trans1	Trans2
1	enet	100001	1500	-	-	-	-	-	0	0

- Vérification des trunks :



Switch1

Physical Config CLI Attributes

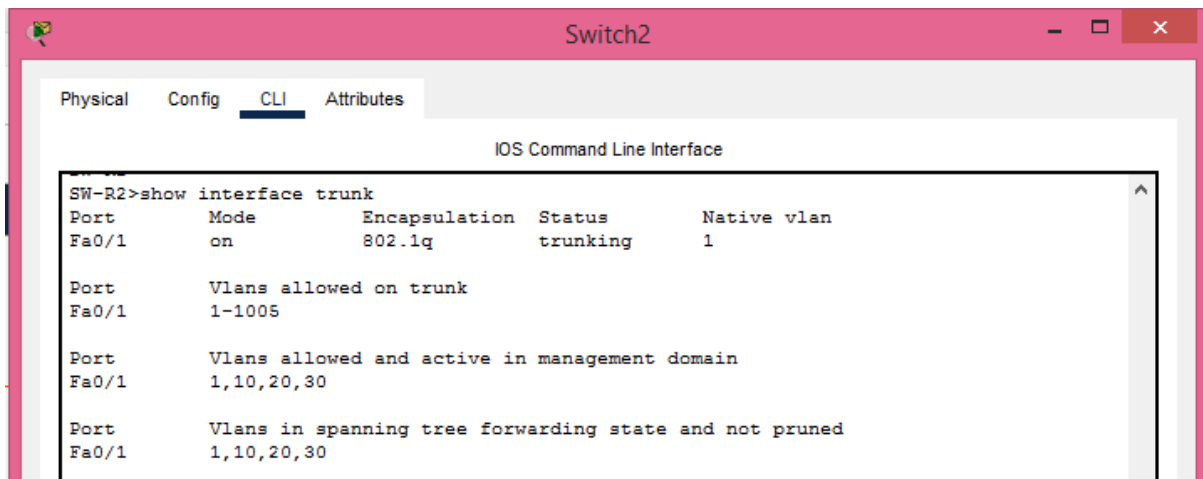
IOS Command Line Interface

```
SW-R2>show interface trunk
Port      Mode      Encapsulation  Status      Native vlan
Fa0/1     on        802.1q         trunking    1

Port      Vlans allowed on trunk
Fa0/1     1-1005

Port      Vlans allowed and active in management domain
Fa0/1     1,10,20,30

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/1     1,10,20,30
```



Switch2

Physical Config CLI Attributes

IOS Command Line Interface

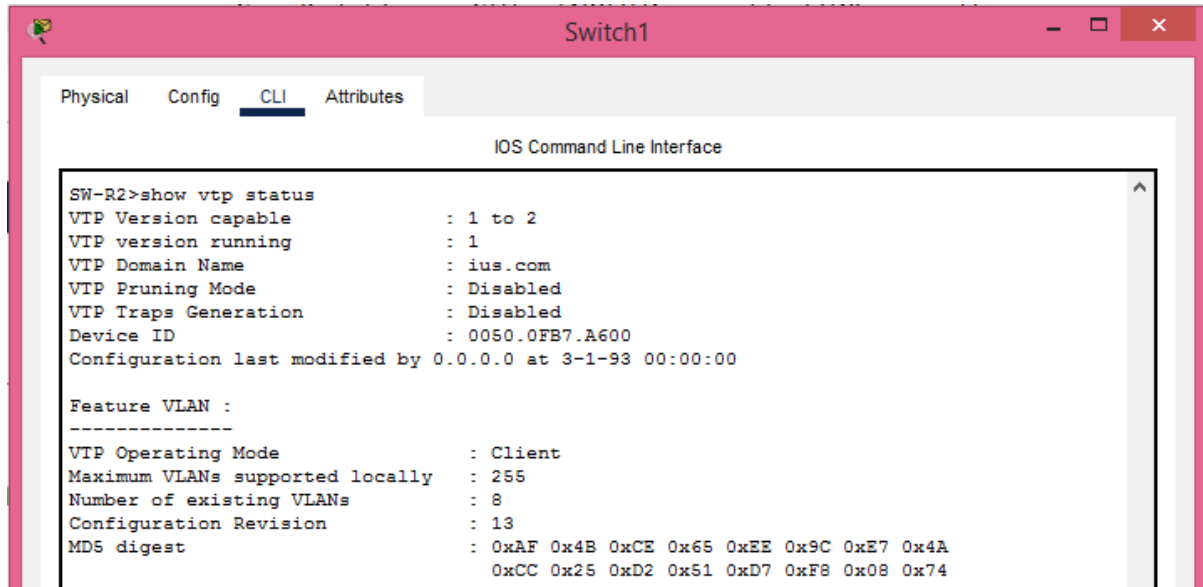
```
SW-R2>show interface trunk
Port      Mode      Encapsulation  Status      Native vlan
Fa0/1     on        802.1q         trunking    1

Port      Vlans allowed on trunk
Fa0/1     1-1005

Port      Vlans allowed and active in management domain
Fa0/1     1,10,20,30

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/1     1,10,20,30
```

- Vérification VTP :



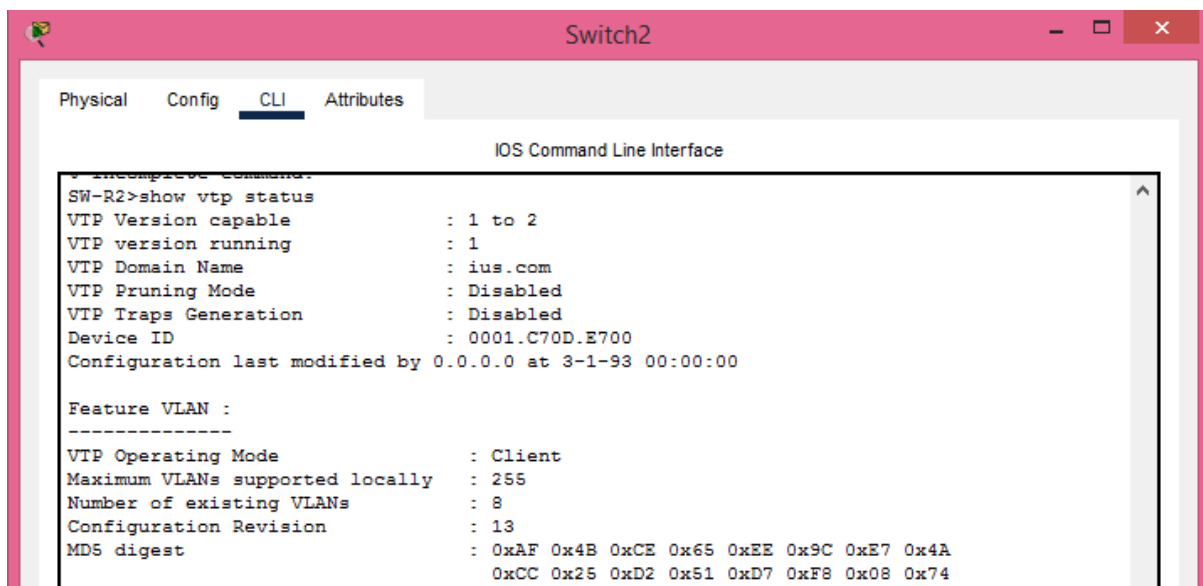
Switch1

Physical Config CLI Attributes

IOS Command Line Interface

```
SW-R2>show vtp status
VTP Version capable      : 1 to 2
VTP version running      : 1
VTP Domain Name          : ius.com
VTP Pruning Mode         : Disabled
VTP Traps Generation     : Disabled
Device ID                : 0050.0FB7.A600
Configuration last modified by 0.0.0.0 at 3-1-93 00:00:00

Feature VLAN :
-----
VTP Operating Mode       : Client
Maximum VLANs supported locally : 255
Number of existing VLANs : 8
Configuration Revision    : 13
MDS digest               : 0xAF 0x4B 0xCE 0x65 0xEE 0x9C 0xE7 0x4A
                          : 0xCC 0x25 0xD2 0x51 0xD7 0xF8 0x08 0x74
```



Switch2

Physical Config CLI Attributes

IOS Command Line Interface

```
SW-R2>show vtp status
VTP Version capable      : 1 to 2
VTP version running      : 1
VTP Domain Name          : ius.com
VTP Pruning Mode         : Disabled
VTP Traps Generation     : Disabled
Device ID                : 0001.C70D.E700
Configuration last modified by 0.0.0.0 at 3-1-93 00:00:00

Feature VLAN :
-----
VTP Operating Mode       : Client
Maximum VLANs supported locally : 255
Number of existing VLANs : 8
Configuration Revision    : 13
MDS digest               : 0xAF 0x4B 0xCE 0x65 0xEE 0x9C 0xE7 0x4A
                          : 0xCC 0x25 0xD2 0x51 0xD7 0xF8 0x08 0x74
```

- Vérification STP :

Switch1

Physical Config CLI Attributes

IOS Command Line Interface

```

4 vlans                0      0      0      8      8

SW-R2>show spanning-tree summary
Switch is in pvst mode
Root bridge for:
Extended system ID      is enabled
Portfast Default        is disabled
PortFast BPDU Guard Default is disabled
Portfast BPDU Filter Default is disabled
Loopguard Default       is disabled
EtherChannel misconfig guard is disabled
UplinkFast              is disabled
BackboneFast            is disabled
Configured Pathcost method used is short

```

Name	Blocking	Listening	Learning	Forwarding	STP Active
VLAN0001	0	0	0	1	1
VLAN0010	0	0	0	1	1
VLAN0020	0	0	0	5	5
VLAN0030	0	0	0	1	1
4 vlans	0	0	0	8	8

Switch2

Physical Config CLI Attributes

IOS Command Line Interface

```

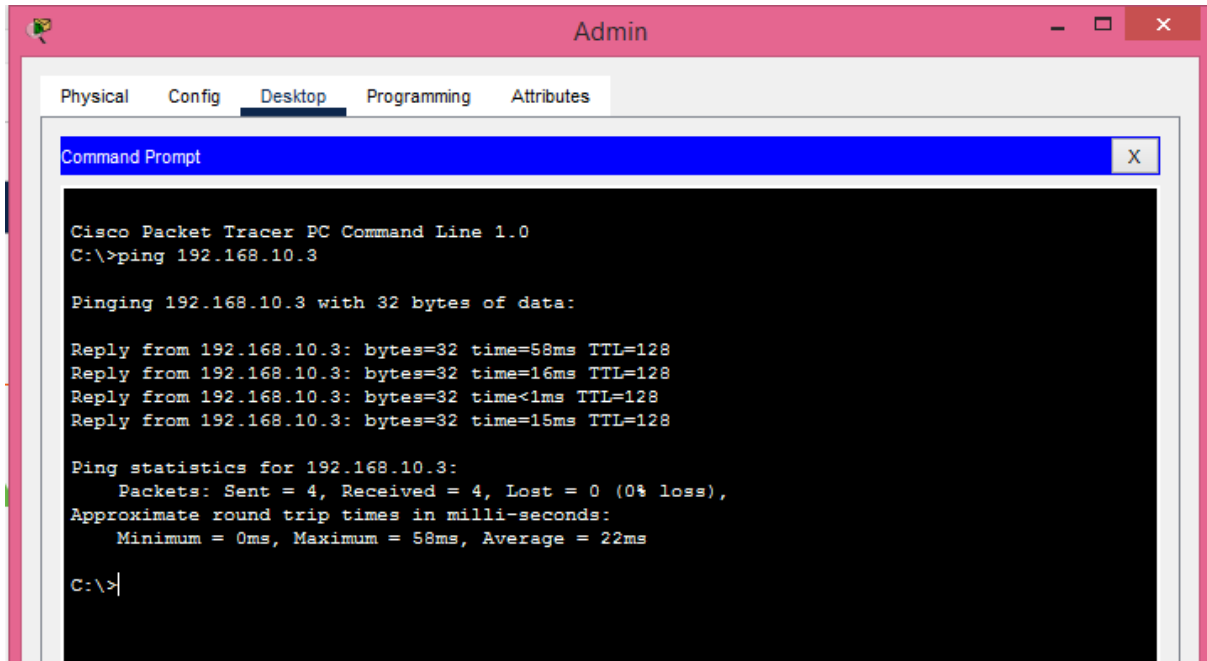
SW-R2>show spanning-tree summary
Switch is in pvst mode
Root bridge for:
Extended system ID      is enabled
Portfast Default        is disabled
PortFast BPDU Guard Default is disabled
Portfast BPDU Filter Default is disabled
Loopguard Default       is disabled
EtherChannel misconfig guard is disabled
UplinkFast              is disabled
BackboneFast            is disabled
Configured Pathcost method used is short

```

Name	Blocking	Listening	Learning	Forwarding	STP Active
VLAN0001	0	0	0	1	1
VLAN0010	0	0	0	2	2
VLAN0020	0	0	0	1	1
VLAN0030	0	0	0	3	3
4 vlans	0	0	0	7	7

Test de connectivite sur :

- VLAN 10 :



The screenshot shows a Cisco Packet Tracer Admin window with a pink title bar and tabs for Physical, Config, Desktop, Programming, and Attributes. The Desktop tab is active, displaying a Command Prompt window. The Command Prompt shows the execution of a ping command to 192.168.10.3, resulting in four successful replies with varying times and a 0% loss rate.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.3

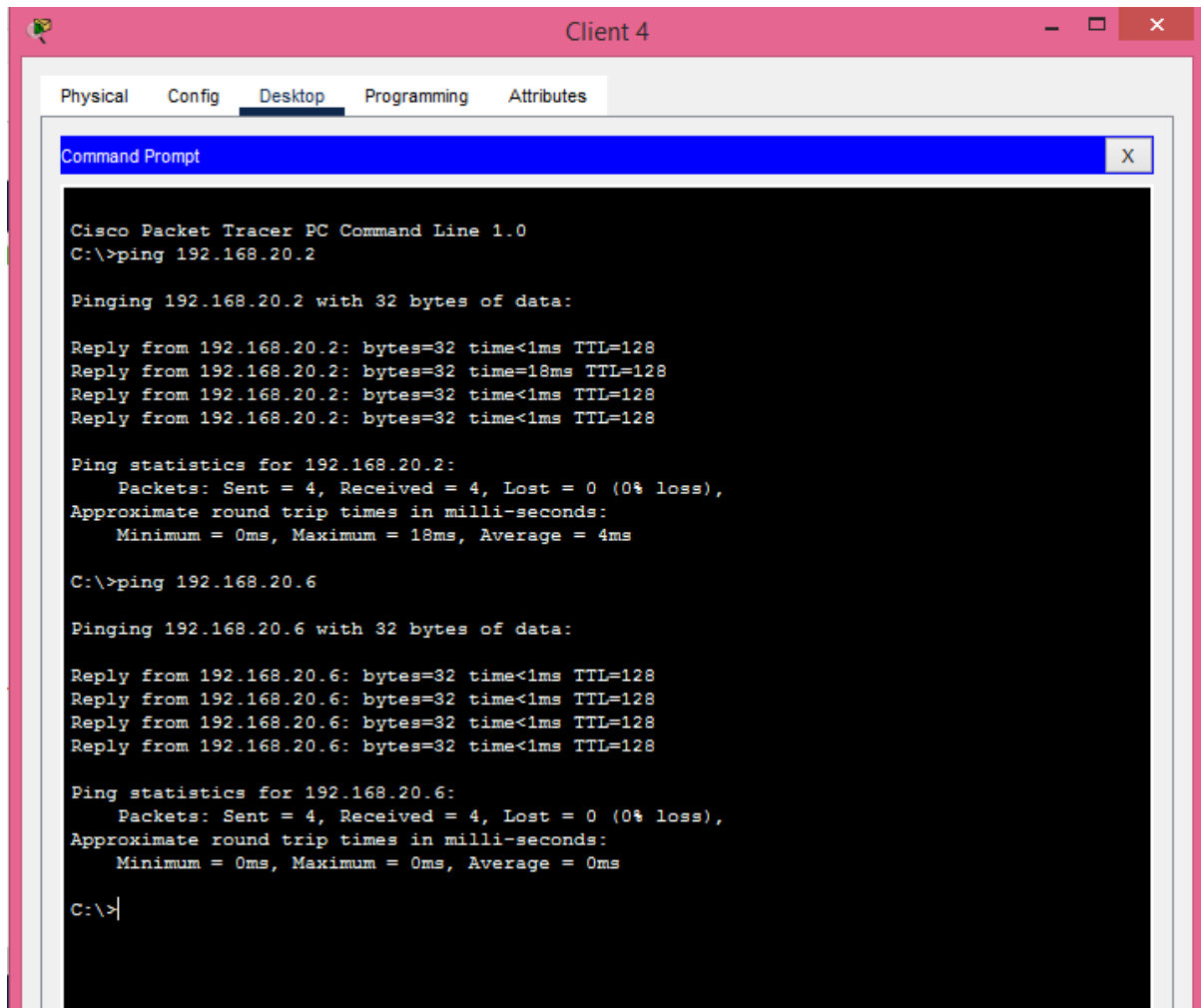
Pinging 192.168.10.3 with 32 bytes of data:

Reply from 192.168.10.3: bytes=32 time=58ms TTL=128
Reply from 192.168.10.3: bytes=32 time=16ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
Reply from 192.168.10.3: bytes=32 time=15ms TTL=128

Ping statistics for 192.168.10.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 58ms, Average = 22ms

C:\>|
```

- VLAN 20 :



- Vlan 30 :

